

Creating a custom grid system similar to Bootstrap

Creating a custom grid system similar to Bootstrap. This grid system will include a **container**, **rows**, and **columns**, and we'll use **media queries** to make it responsive for different screen sizes.

Below is the CSS code for creating a responsive grid system. This includes a `.container` to center content, `.row` to hold the columns, and multiple `.col` classes to define column sizes.

Example HTML Structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Custom Grid System</title>
<link rel="stylesheet" href="grid.css">
</head>
<body>
  <div class="container">
    <div class="row">
      <div class="col col-6 col-md-4 col-lg-3">Column 1</div>
      <div class="col col-6 col-md-4 col-lg-3">Column 2</div>
      <div class="col col-6 col-md-4 col-lg-3">Column 3</div>
      <div class="col col-6 col-md-4 col-lg-3">Column 4</div>
    </div>
  </div>
</body>
</html>
```

CSS for Grid System (`grid.css`):

```
/* Base Styles */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}
```

```
.container {
  width: 100%;
  max-width: 1200px;
  margin: 0 auto; /* Center the container */
  padding: 0 15px; /* Add some padding */
}

/* Row Styles */
.row {
  display: flex;
  flex-wrap: wrap; /* Allow columns to wrap to the next line if necessary */
  margin: -15px; /* Offset column padding to align content */
}

/* Column Styles */
.col {
  padding: 15px; /* Add padding to create spacing between columns */
}

/* Mobile First Default Columns */
.col-1 { width: 8.33%; }
.col-2 { width: 16.66%; }
.col-3 { width: 25%; }
.col-4 { width: 33.33%; }
.col-5 { width: 41.66%; }
.col-6 { width: 50%; }
.col-7 { width: 58.33%; }
.col-8 { width: 66.66%; }
.col-9 { width: 75%; }
.col-10 { width: 83.33%; }
.col-11 { width: 91.66%; }
.col-12 { width: 100%; }

/* Medium Screens (Tablets) */
@media (min-width: 768px) {
  .col-md-1 { width: 8.33%; }
  .col-md-2 { width: 16.66%; }
  .col-md-3 { width: 25%; }
  .col-md-4 { width: 33.33%; }
  .col-md-5 { width: 41.66%; }
  .col-md-6 { width: 50%; }
  .col-md-7 { width: 58.33%; }
  .col-md-8 { width: 66.66%; }
  .col-md-9 { width: 75%; }
  .col-md-10 { width: 83.33%; }
}
```

```

.col-md-11 { width: 91.66%; }
.col-md-12 { width: 100%; }
}

/* Large Screens (Desktops) */
@media (min-width: 992px) {
.col-lg-1 { width: 8.33%; }
.col-lg-2 { width: 16.66%; }
.col-lg-3 { width: 25%; }
.col-lg-4 { width: 33.33%; }
.col-lg-5 { width: 41.66%; }
.col-lg-6 { width: 50%; }
.col-lg-7 { width: 58.33%; }
.col-lg-8 { width: 66.66%; }
.col-lg-9 { width: 75%; }
.col-lg-10 { width: 83.33%; }
.col-lg-11 { width: 91.66%; }
.col-lg-12 { width: 100%; }
}

```

Explanation:

1. Container (`.container`):

- This element centers the content, provides a max-width of `1200px` , and adds side padding.

2. Row (`.row`):

- The `.row` uses `display: flex` to align columns horizontally.
- `flex-wrap: wrap` makes the columns wrap to the next line if there's not enough space.
- Negative margin (`-15px`) helps to adjust for the column's inner padding (`15px`), keeping everything neatly aligned.

3. Column (`.col`):

- **Base Classes:**

- Columns have padding (`15px`) to provide space between columns.
- You can use classes like `.col-6` for 50% width, `.col-4` for 33.33%, etc.

- **Responsive Classes:**

- The media queries define classes for different screen sizes:
 - `.col-md-*` applies to medium screens (`min-width: 768px`) such as tablets.
 - `.col-lg-*` applies to large screens (`min-width: 992px`) such as desktops.

- The column widths adjust depending on the screen size, creating a responsive grid system.

Usage:

- **Mobile-First Approach:** By default, the columns take on their base width (`col*`), suitable for mobile devices.
- **Responsive Adjustments:** For medium screens and larger, you can use `.col-md*` or `.col-lg*` to specify different column sizes based on the device.

For instance:

- **Mobile (default):** Each column takes 50% of the width (`.col-6`).
- **Tablet:** On tablets, each column can take 33.33% (`.col-md-4`), resulting in three columns per row.
- **Desktop:** On desktops, each column can take 25% (`.col-lg-3`), resulting in four columns per row.